Grade 12 Advanced Functions
Rate of Change Test

First Name: _____

Last Name: _____

Directions:

- Please answer to 2 decimal points
- The test is designed to be completed in 75 minutes

For grading use only

Page:	2	3	4	5	Total
Points:	5	18	9	6	38
Score:					

Multiple Choice (10 marks)

1. (1 point) The average rate of change (AROC) of $f(x) = x^2 - 4x$ on $[1, 5]$ is:
   A. 1
   B. 2
   C. 4
   D. -2

2. (1 point) The instantaneous rate of change (IROC) of f at $x = a$ is equal to:
   A. $\dfrac{f(a+1) - f(a)}{1}$
   B. $\lim\limits_{h \to 0} \dfrac{f(a+h) - f(a)}{h}$ (Newton's quotient limit)
   C. $\dfrac{f(a) - f(0)}{a - 0}$
   D. $\dfrac{f(a+h) - f(a-h)}{2h}$ for any fixed h

3. (1 point) AROC over $[a, b]$ is the slope of the:
   A. Tangent line at $x = a$
   B. Tangent line at $x = b$
   C. Secant line through $(a, f(a))$ and $(b, f(b))$
   D. Horizontal line $y = f\left(\dfrac{a+b}{2}\right)$

4. (1 point) Evaluate the limit: $\lim\limits_{x \to 3} \dfrac{x^2 - 9}{x - 3}$.
   A. 3
   B. 6
   C. 9
   D. Does not exist

5. (1 point) For $f(x) = \sqrt{x+1}$, the IROC at $x = 3$ equals (using the limit of the difference quotient):
   A. $\dfrac{1}{4}$
   B. $\dfrac{1}{8}$
   C. $\dfrac{1}{2}$
   D. 1

Long Answer (33 marks total)

6. AROC and IROC via Newton's Quotient (10 marks) Let $f(x) = x^2 - 4x + 1$.
   (a) (4 points) Compute the average rate of change of f on $[1, 1+h]$ in terms of h (simplify fully).
   (b) (4 points) Use your result to find the instantaneous rate of change at $x = 1$ by taking the limit as $h \to 0$ (show every algebraic step).
   (c) (2 points) Interpret your numerical result from part (b) if f measures position (meters) and x is time (seconds).

7. AROC and IROC from a Table (8 marks) A particle's position (meters) at times t (seconds) is:

t (s)	0	1	2	3	4	5
$s(t)$ (m)	0.0	1.1	3.8	7.7	12.5	18.0

   (a) (3 points) Compute the average rate of change on $[1, 4]$ with proper units.
   (b) (3 points) Estimate the instantaneous rate of change at $t = 3$ using a symmetric secant around $t = 3$.
   (c) (2 points) Briefly interpret both answers (speeding up, slowing down, or roughly constant?).

8. Limits and IROC by Algebraic Simplification (9 marks) Consider $f(x) = \sqrt{x+1}$.
   (a) (5 points) Compute the instantaneous rate of change at $x = 3$ using the limit of the difference quotient. Do not use rules; rationalize carefully.
   (b) (2 points) Write the AROC over $[3, 3+h]$ in simplest form (in terms of h), showing the algebra.
   (c) (2 points) Evaluate your AROC formula for $h = 0.1$ and $h = -0.1$ and compare to your IROC value.

9. Population of a Bacterial Colony using IROC (6 marks) A bacterial colony's population (cells) is modeled by $P(t) = t^2 + 6t + 200$, where t is time in hours.
   (a) (3 points) Compute the average rate of change of P on $[4, 7]$. Include units and interpretation.
   (b) (3 points) Find the instantaneous rate of change at $t = 5$ using the limit of the Newton's quotient (no rules). Interpret in context.